Replication of Card & Krueger (1994): Minimum Wages and Employment

A Difference-in-Differences Analysis of the New Jersey Fast-Food Industry

YANG Ruochen · Student ID: 3036770041

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1. Research question and motivation

Question. Did New Jersey's April 1, 1992 increase in the minimum wage from $4.25 to $5.05 per hour reduce employment in fast-food restaurants?

Motivation. The paper is the canonical application of the difference-in-differences (DiD) design and the most influential empirical test of the competitive theory of the labour market. A textbook competitive model predicts that a binding minimum wage reduces employment along the labour-demand curve. Card and Krueger exploit a natural experiment: New Jersey (NJ) raised its minimum wage while neighbouring eastern Pennsylvania (PA), across the Delaware River, left its wage floor unchanged at $4.25. The two states share a regional labour market, making PA a credible comparison group.

Hypotheses. Under the competitive benchmark the NJ–PA difference in employment growth should be negative. Under monopsony or search-friction models, employment may be unchanged or even rise. The paper's headline finding — no evidence of an employment decline, and if anything a relative increase — is the result we set out to reproduce.

2. Data sources, sample construction, and variable definitions

Source. Public replication data from Card & Krueger (1994), distributed as the Princeton restaurant survey (`public.csv`). The file contains 410 store-level observations (46 columns): 331 stores in New Jersey (`STATE = 1`) and 79 in eastern Pennsylvania (`STATE = 0`), from four chains — Burger King (171), KFC (80), Roy Rogers (99), and Wendy's (60). Stores were surveyed twice: wave 1 in February–March 1992 (before the increase) and wave 2 in November–December 1992 (after).

Sample construction. We follow the paper's sample definitions. For the descriptive means (Table 2) and the level/change tabulations (Table 3), each cell uses all stores with non-missing data for the relevant variable; wave-2 employment is set to zero for the six permanently closed stores and

treated as missing for the four temporarily closed stores and one refusal. The *balanced subsample* (Table 3, row 4) consists of the 384 stores with employment in both waves (309 NJ, 75 PA). The *regression sample* (Table 4) further requires a valid wave-1 starting wage (needed to construct the wage gap), leaving 365 stores; the paper's analogous sample is 357. We do not publish the raw data; see `README.md` for access instructions.

Variable definitions.

Variable	Definition
FTE employment	`EMPFT + NMGRS + 0.5 * EMPPT` (full-time + managers + half of part-time)
Headcount	`EMPFT + EMPPT + NMGRS`
`NJ` (treatment)	1 if `STATE == 1`, 0 if Pennsylvania
Change in FTE, `dfte`	wave-2 FTE minus wave-1 FTE
Wage gap, `GAP`	`(5.05 - W1)/W1` for NJ stores with wave-1 wage `W1 < 5.05`; 0 otherwise
% full-time	`100 * EMPFT / FTE` (full-time, excluding managers, as a share of FTE)
Full-meal price	`PSODA + PFRY + PENTREE` (medium soda + small fries + entree, incl. tax)
Controls	chain dummies (`CHAIN`), company-owned (`CO_OWNED`), NJ sub-region dummies (`CENTRALJ`, `SOUTHJ`, with north NJ omitted) and PA sub-region dummies (`PA1`, `PA2`)

Cleaning log. All cleaning is coded in `src/prep.py` and reproduced by `run_all.py`. Missing starting wages are kept as a separate category where the paper reports wage distributions; employment outcomes are computed only from non-missing components.

3. Empirical design

The DiD estimand is

$$\delta = (NJ_{after} - NJ_{before}) - (PA_{after} - PA_{before})$$

We estimate it by ordinary least squares on first differences,

$$\Delta FTE_i = \alpha + \delta \cdot NJ_i + \varepsilon_i$$

which is numerically identical, in a balanced panel, to the two-way fixed-effects specification $FTE_{it} = \beta \cdot NJ_i + \gamma \cdot After_t + \delta \cdot (NJ_i \times After_t) + \varepsilon_{it}$. Standard errors are conventional (homoskedastic) OLS, matching the paper's Table 4. We then add controls successively: chain and ownership dummies, then the wage gap (`GAP`) as a treatment-intensity variable, then regional dummies.

Identification. The identifying assumption is parallel trends: absent the wage increase, employment in NJ and PA would have evolved identically. With only two waves the assumption cannot be tested directly; we support it by (i) the regional proximity of the two states, (ii) a compliance check showing the wage increase was actually implemented (Table 2), and (iii) robustness of the estimate to controls and sample choices (Table 6). Within New Jersey, comparing low-wage stores (which should be affected) with high-wage stores (which should not) provides a further specification test, and the paper shows this comparison is passed.

4. Core replication results and comparison with the original paper

Table 2 — descriptive means. The table reproduces the paper's Table 2. Wave-1 starting wages average \$4.61 in NJ and \$4.63 in PA, with 30.5% of NJ stores and 32.9% of PA stores paying exactly the old \$4.25 minimum. After the increase, the NJ mean wage rises to \$5.08 while PA is unchanged at \$4.62; 85.5% of NJ stores report a starting wage of exactly \$5.05 in wave 2, versus 1.3% in PA. The treatment was implemented as intended (Figure 1). Employment in wave 1 averaged 20.4 FTE in NJ and 23.3 in PA; the fraction full-time was 32.9% (NJ) and 35.0% (PA), rising to 35.9% and falling to 30.4% in wave 2 — the two variables the paper flags as showing a relative NJ increase besides wages. The full-meal price rose from \$3.35 to \$3.41 in NJ but was flat at \$3.04 in PA.

Table 3 — average employment and the main DiD.

FTE employment	PA	NJ	NJ−PA	NJ: \$4.25	NJ: \$4.26–4.99	NJ: ≥\$5.00	Low−High	Mid−High
Before, all available	23.33	20.44	−2.89	19.56	20.08	22.25	−2.69	−2.17
After, all available	21.17	21.03	−0.14	20.88	20.96	20.21	+0.66	+0.74
Change in mean FTE	−2.17	+0.59	**+2.75**	+1.32	+0.87	−2.04	+3.36	+2.91

FTE employment	PA	NJ	NJ−PA	NJ: $4.25	NJ: $4.26–4.99	NJ: ≥$5.00	Low−High	Mid−High
Change, balanced sample	−2.28	+0.47	**+2.75**	+1.20	+0.71	−2.16	+3.36	+2.87
Change, temp-closed = 0	−2.28	+0.23	**+2.51**	+0.90	+0.49	−2.39	+3.29	+2.88

FTE employment rose in NJ and fell in PA, so the DiD estimate is positive: **+2.75 FTE workers**, essentially identical to the paper's headline +2.76. Our balanced first-difference regression gives a conventional standard error of 1.15 (t = 2.38). The estimate is unchanged when the sample is restricted to the balanced subsample (row 4) and only slightly smaller when wave-2 employment at the temporarily closed stores is set to zero (row 5, +2.51). Within New Jersey, employment expanded at the low-wage stores (paying $4.25 in wave 1, +1.20 balanced) and contracted at the high-wage stores (≥$5.00, −2.16 balanced) — the high-wage change is almost identical to the PA change of −2.28, exactly as the paper reports, providing a within-state specification test.

Table 4 — reduced-form models for the change in employment. The dependent variable is the change in FTE employment; sample N = 365 (the paper uses 357).

Model	Coefficient	s.e.	t	N
(i) NJ dummy	+2.36	1.15	2.06	365
(ii) NJ + chain & ownership	+2.33	1.15	2.02	365
(iii) GAP	+15.89	5.95	2.67	365
(iv) GAP + chain & ownership	+15.19	6.08	2.50	365
(v) GAP + chain & ownership + region	+12.03	7.32	1.64	365

The NJ dummy is positive and stable across specifications (2.36 and 2.33). The `GAP` coefficient implies that a store initially paying the old $4.25 minimum (GAP = 0.188) gained about 0.188 × 15.89 ≈ 3.0 FTE workers relative to an unaffected store. Adding the regional dummies attenuates the GAP coefficient to 12.03 and raises its standard error, so that column (v) is no

longer significant at the 5% level — precisely the pattern the paper reports and attributes to measurement error in the starting wage.

Comparison with the original paper. Every point estimate is reproduced to within sampling noise:

Estimate	Ours	Paper
Table 3 DiD (change in mean FTE)	+2.75	+2.76 (t = 2.03)
Table 4 (i) NJ dummy	+2.36 (1.15)	+2.33 (1.19)
Table 4 (iii) GAP	+15.89 (5.95)	+15.65 (6.08)
Table 4 (v) GAP + region	+12.03 (7.32)	+11.91 (7.39)

The small residual differences are driven by the sample: our Table 4 regression sample has 365 stores against the paper's 357, reflecting a few stores whose wave-2 starting wage is missing in the public file. The headline conclusion — no employment decline, and a positive point estimate — is reproduced exactly.

5. Independent extension: menu-price pass-through

Question. The minimum wage raised the cost of labour. Did restaurants pass any of that cost through to consumers via higher menu prices? The original paper discusses prices descriptively but does not estimate a formal DiD for them.

Design. We apply the same DiD design to the change in each of the three menu prices recorded in both waves (medium soda, small fries, entree), with chain and ownership controls:

$$\Delta Price_i = \alpha + \delta \cdot NJ_i + X_i'\gamma + \varepsilon_i$$

Results.

Menu item	NJ mean change	PA mean change	DiD	s.e.	t	N
Medium soda	+0.000	+0.008	−0.005	0.008	−0.62	380
Small fries	+0.017	+0.019	−0.001	0.009	−0.10	366
Entree	+0.043	−0.033	**+0.088**	0.045	1.94	375

Interpretation. Soda and fries show no detectable pass-through. The entree — the highest-priced item — rose by about 8.8 cents in NJ relative to PA (t = 1.94), roughly 6.5% of the ~$1.35 entree price. This is consistent with *partial, item-specific* pass-through: restaurants absorbed most of the wage increase (in line with the finding that employment did not fall) while adjusting a single higher-margin menu item. The result is economically modest relative to the $0.80 wage increase, indicating that most of the cost was absorbed through other margins (labour demand, hours, or margins).

6. Validation, limitations, and conclusion

Validation. (i) The compliance check (Table 2) confirms the treatment was binding and implemented. (ii) The main DiD is stable across controls (Table 4) and sample choices (Table 6): +2.75 for FTE, +2.41 for headcount, and +2.42 when closed stores are treated as having zero wave-2 employment. Restricting to "compliers" — stores paying exactly the old $4.25 minimum before the increase — *strengthens* the result to +3.22 (t = 1.81), consistent with a treatment-intensity story: the most-treated stores show the largest relative employment gain. The full-time/part-time split (+0.69 full-time, −1.36 part-time, both insignificant) suggests the relative gain was concentrated among full-time workers.

Limitations. (i) With only two waves, the parallel-trends assumption cannot be tested; the design relies on the regional comparison. (ii) Treatment is assigned by state, not at random, and the early 1990s recession may have hit the two states differently. (iii) Some NJ stores already paid ≥ $5.05 before the increase, attenuating the measured effect. (iv) Conventional standard errors do not address spatial correlation across stores within a labour market. (v) Survey non-response and store closures are potential sources of selection, which we address but cannot fully rule out.

Conclusion. The replication reproduces Card & Krueger's central finding: New Jersey's minimum-wage increase did not reduce fast-food employment relative to Pennsylvania; the DiD point estimate is positive (+2.75 FTE, matching the paper's +2.76). The extension adds a new, testable margin — menu-price pass-through — and finds modest, item-specific pass-through concentrated in the entree, consistent with firms absorbing the cost largely through margins rather than employment or prices.

References

Card, David, and Alan B. Krueger. 1994. "Minimum Wages and Employment: A Case Study of the Fast-Food Industry in New Jersey and Pennsylvania." *American Economic Review* 84(4): 772–793.